THE MASTERMIND COACHING CLASSES

Revision Notes

Class 6 · Mathematics

Ganita Prakash — Part 2, Chapters 6 to 10 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
6	Perimeter and Area	Perimeter and area of rectangles, squares, triangles, parallelograms; units; irregular shapes
7	Fractions	Kinds of fractions, equivalent fractions, comparing, adding and subtracting, mixed fractions
8	Playing with Constructions	Circles and compass work, rectangles and squares, diagonals, perpendicular bisector, equidistant points
9	Symmetry	Line symmetry, reflection, rotational symmetry, order and angle of rotation
10	The Other Side of Zero	Integers, number line, comparing, addition and subtraction of integers, real-life uses

The Mastermind Coaching Classes — Road No. 5, VKIA, Jaipur — 9887258679

CHAPTER 06 · MATHS (PART 2)

Perimeter and Area

Fencing a field needs perimeter; tiling a floor needs area. This chapter defines both, derives the formulae for rectangles, squares, triangles and parallelograms by cutting and rearranging, converts units, and estimates the area of irregular shapes by counting squares.

Big Questions

- How do perimeter and area differ and how are they measured?
- How do we get the area of a triangle and a parallelogram from a rectangle?
- How can the area of an irregular shape be estimated?

6.1 Perimeter

DEFINITION

Perimeter is the total length of the boundary of a closed figure. It is measured in units of length (cm, m, km).

Figure	Perimeter	Example
Rectangle (l, b)	**2 × (l + b)**	25 m × 18 m: 2 × 43 = 86 m
Square (side a)	**4a**	a = 12 cm: 48 cm
Equilateral triangle (a)	**3a**	a = 13 cm: 39 cm
Regular polygon of n sides	**n × side**	regular pentagon, side 6 cm: 30 cm
Any polygon	sum of all sides	

Reverse problems. A square has perimeter 48 cm, so side = 48 ÷ 4 = **12 cm**. A rectangle has perimeter 60 m and length 18 m, so breadth = 60 ÷ 2 − 18 = **12 m**.

6.2 Area

DEFINITION

Area is the measure of the surface enclosed by a closed figure. It is measured in **square units** (cm^2, m^2, km^2).

Figure	Area	Example
Rectangle (l, b)	**l × b**	25 m × 18 m = 450 m^2
Square (side a)	**a × a = a^2**	side 11 m: 121 m^2
Parallelogram	**base × height**	base 10 cm, height 6 cm: 60 cm^2
Triangle	**½ × base × height**	base 9 cm, height 4 cm: 18 cm^2

> **REMEMBER**
> The **height** is the perpendicular distance from the base to the opposite side or vertex, **not** the slanting side. A diagonal splits a rectangle into **two congruent triangles**, each half the area.

▸ **How the formulae arise**

- **Parallelogram:** cut a right-angled triangle from one end and slide it to the other end; you get a rectangle with the same base and height.
- **Triangle:** two copies of the same triangle fit together to form a parallelogram, so the triangle is **half** of base × height.
- **Reverse:** a triangle with area 54 cm^2 and base 12 cm has height 2 × 54 ÷ 12 = **9 cm**.

6.3 Units of Area

Conversion	Why
1 m^2 = 10,000 cm^2	1 m = 100 cm, so 100 × 100
1 cm^2 = 100 mm^2	1 cm = 10 mm
1 km^2 = 10,00,000 m^2	1 km = 1000 m
1 hectare = 10,000 m^2	land measure

3 m^2 = 3 × 10,000 = **30,000 cm^2**; 50,000 cm^2 = 50,000 ÷ 10,000 = **5 m^2**; 2.5 m^2 = **25,000 cm^2**. Always change lengths to the same unit **before** multiplying.

6.4 Combined and Irregular Figures

Composite figures: split into rectangles and triangles, find each area and add; or find the area of the big rectangle and subtract the missing part. An L-shape formed by a 10 × 6 rectangle with a 4 × 2 corner removed has area 60 − 8 = **52 square units**.

Step 1
Place the shape on squared paper.

Step 2
Count the **full** squares.

Step 3
Count the squares that are **more than half** covered as 1, the ones **about half** as ½, and ignore the smaller bits.

Step 4
Add; the total is an **estimate** of the area in square units.

> **NOTE**
> Shapes with the **same perimeter** can have **different areas**. A rectangle of area 36 cm^2 can have sides (1, 36), (2, 18), (3, 12), (4, 9) or (6, 6) with perimeters 74, 40, 30, 26, 24 cm. For a fixed perimeter the **square** has the greatest area.

6.5 Word Problems

Fencing and tiling. A 25 m × 18 m garden needs 86 m of wire for a fence (perimeter) and 450 m^2 of turf (area). A path of width 1 m round a square park of side 40 m: **outer square 42 m**, area of the path = $42^2 - 40^2$ = 1764 − 1600 = **164 m^2**.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Define perimeter and area.	Perimeter is the length of the boundary; area is the surface enclosed.
Area of a square of side 11 m?	121 m^2.
Perimeter of a regular pentagon of side 6 cm?	30 cm.
Convert 3 m^2 into cm^2.	30,000 cm^2.
Area of a triangle of base 9 cm and height 4 cm?	½ × 9 × 4 = 18 cm^2.
How does a diagonal divide a rectangle?	Into two congruent triangles, each half the area.
Square of area 169 m^2: side and perimeter?	13 m and 52 m.
Area of a parallelogram of base 10 cm and height 6 cm?	60 cm^2.
How do you find the area of an irregular shape?	Count full squares, treat part squares by the half rule and add.
Do shapes with equal perimeter have equal area?	Not necessarily; e.g. 6 × 6 and 4 × 8 both have perimeter 24 but areas 36 and 32.

★ KEY TAKEAWAYS

- **Perimeter** = boundary length; **area** = surface enclosed (square units).
- Rectangle: P = 2(l + b), A = l × b; square: P = 4a, A = a^2.
- Parallelogram = base × height; **triangle = ½ × base × height**.
- 1 m^2 = 10,000 cm^2; convert lengths to one unit before multiplying.
- Same perimeter does **not** mean same area; irregular areas are estimated by **counting squares**.

CHAPTER 07 · MATHS (PART 2)

Fractions

A fraction names a part of a whole or a share of a collection. This chapter covers the meaning and kinds of fractions, equivalent fractions and the lowest form, comparing fractions, converting between mixed and improper forms, and adding and subtracting fractions in word problems.

Big Questions

- What does a fraction mean and what kinds are there?
- How do we compare fractions and find equivalent ones?
- How do we add and subtract fractions with unlike denominators?

7.1 Meaning of a Fraction

DEFINITION

A **fraction** a/b has a **numerator** a (parts taken) and a **denominator** b (equal parts in the whole), with b ≠ 0. It also means **a ÷ b**: sharing 5 chapatis equally among 4 children gives each 5/4 = 1 1/4 chapati.

Kind	Description	Examples
Proper fraction	numerator < denominator (less than 1)	3/5, 7/10
Improper fraction	numerator ≥ denominator (1 or more)	19/6, 7/7
Mixed fraction	whole number + proper fraction	3 1/6, 2 3/4
Unit fraction	numerator 1	1/2, 1/7
Like fractions	same denominator	2/9 and 5/9
Unlike fractions	different denominators	1/3 and 1/4

Fractions are **numbers** on the number line: 0, 1/4, 1/2, 3/4, 1, 1 1/4, … The fraction of an hour for 15 minutes is 15/60 = 1/4; the fraction of a metre for 35 cm is 35/100 = 7/20.

7.2 Equivalent Fractions and Lowest Form

FORMULA

Multiply or divide **both** numerator and denominator by the same non-zero number to get an **equivalent fraction**: a/b = (a × k)/(b × k) = (a ÷ k)/(b ÷ k).

- 2/3 = 4/6 = 6/9 = 8/12 (all show the same part of the whole).
- A fraction is in **lowest (simplest) form** when numerator and denominator have **no common factor** except 1: divide both by their HCF. 24/36 ÷ 12 = **2/3**.
- Two fractions a/b and c/d are equal if **a × d = b × c** (cross-multiplication): 2/3 and 8/12 → 2 × 12 = 24 = 3 × 8.

7.3 Mixed and Improper Fractions

Conversion	Method	Example
Mixed → improper	(whole × denominator + numerator) / denominator	5 3/8 = (5 × 8 + 3)/8 = 43/8
Improper → mixed	divide numerator by denominator: quotient = whole part, remainder = new numerator	37/5 = 7 2/5

7.4 Comparing Fractions

Same denominator
The larger numerator is the larger fraction: 5/9 > 2/9.

Same numerator
The smaller denominator is the larger fraction: 3/4 > 3/7 (fewer, bigger pieces).

Different both
Convert to **like fractions** using the LCM of the denominators, then compare numerators.

Example. Compare 5/8 and 7/12: LCM 24, so 15/24 and 14/24; hence **5/8 > 7/12**. Arrange 2/3, 3/5, 7/10 in descending order: LCM 30 gives 20/30, 18/30, 21/30, so **7/10 > 2/3 > 3/5**.

7.5 Adding and Subtracting Fractions

FORMULA

Like fractions: a/c ± b/c = (a ± b)/c. **Unlike fractions:** first make the denominators equal (use the LCM), then add or subtract the numerators and simplify.

2/3 + 3/4: LCM 12, so 8/12 + 9/12 = 17/12 = **1 5/12**. **7/9 − 2/9 = 5/9. 1/2 + 1/4 + 1/8 = 4/8 + 2/8 + 1/8 = 7/8.**

Mixed fractions: convert to improper fractions or subtract whole and fraction parts separately. 3 1/4 − 1 1/2 = 13/4 − 6/4 = 7/4 = **1 3/4**.

▸ Word problems

- Ravi ate 2/5 and Rita 1/3 of a cake: together 6/15 + 5/15 = **11/15**; left = **4/15**.
- A cloth of 7/2 m has a piece of 5/4 m cut off: left = 14/4 − 5/4 = **9/4 m = 2 1/4 m**.
- A bag has 24 balls, 3/8 are red: 3/8 of 24 = 9 red and 15 not red.

REMEMBER

Never add denominators: 1/2 + 1/3 is **not** 2/5. The denominator tells the **size of the piece**, and pieces can be added only when their sizes match.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Define proper and improper fractions with an example each.	Proper: numerator < denominator, 3/5. Improper: numerator ≥ denominator, 19/6.
Convert 19/6 to a mixed fraction.	3 1/6.
Simplify 24/36.	2/3 (divide by 12).

Likely question	Model answer
What is a unit fraction?	A fraction with numerator 1, such as 1/7.
What fraction of an hour is 15 minutes?	1/4.
Add 2/3 and 3/4.	17/12 = 1 5/12.
Compare 5/8 and 7/12.	15/24 > 14/24, so 5/8 is greater.
Write 5 3/8 as an improper fraction.	43/8.
Subtract 3 1/4 − 1 1/2.	1 3/4.
Share 5 equal chapatis among 4 children.	Each gets 5/4 = 1 1/4 chapati.

★ KEY TAKEAWAYS

- A fraction a/b means **a out of b equal parts** or **a ÷ b**.
- **Proper, improper, mixed, unit, like, unlike** fractions: know each with an example.
- Equivalent fractions: multiply or divide **numerator and denominator** by the same number; lowest form uses the **HCF**.
- To compare or add unlike fractions, make **like fractions** with the **LCM**.
- Add or subtract the **numerators only**; the denominator stays the same.

CHAPTER 08 · MATHS (PART 2)

Playing with Constructions

Constructions are drawings made with exact tools: a ruler, a compass and a set square. This chapter starts with circles and their parts, builds rectangles and squares from their properties, and constructs the perpendicular bisector of a segment and the set of points equidistant from two points.

Big Questions

- Which parts of a circle can a compass draw and what are they called?
- How do we construct a rectangle or square and check it?
- How do we draw a perpendicular bisector and what does it tell us?

8.1 Tools and the Circle

The **ruler** draws straight lines and measures length; the **compass** draws circles and arcs and copies lengths; the **set squares** draw right angles and parallel lines; the **divider** copies lengths; the **protractor** measures angles.

Part of a circle	Meaning
Centre	fixed point, equally far from every point of the circle
Radius	segment from the centre to a point on the circle
Diameter	chord through the centre; = 2 × radius; the longest chord
Chord	segment joining two points of the circle
Arc	a part of the circle (circumference)
Concentric circles	circles with the same centre but different radii

NOTE

All points of a circle are at the same distance (radius) from the centre. A compass **opening** equals the radius. To copy a segment AB, set the compass to AB and mark the same length on a ray.

8.2 Rectangles and Squares

Property	Rectangle	Square
Opposite sides	equal and parallel	equal and parallel
All sides	not necessarily equal	**all four equal**
Angles	each 90° (4 right angles)	each 90°
Diagonals	**equal**; bisect each other	**equal**; bisect each other **at right angles**
Lines of symmetry	2	4

Every square is a rectangle (it has all rectangle properties) but **not every rectangle is a square** (sides may be unequal). A quadrilateral whose diagonals are equal and bisect each other is a rectangle; if they also meet

at 90° it is a square.

Construct a 6 cm × 4 cm rectangle
Draw base AB = 6 cm. At A and B draw perpendiculars (set square) and mark AD = BC = 4 cm. Join D and C; check DC = 6 cm and the diagonals are equal.

Construct a square of side 5 cm
Draw AB = 5 cm, raise perpendiculars at A and B, cut off AD = BC = 5 cm, join D and C; check the diagonals are equal and meet at right angles.

8.3 Perpendicular Bisector and Equidistant Points

DEFINITION

The **perpendicular bisector** of a segment AB is a line that cuts AB at its **midpoint** and is **perpendicular** to it. Every point on it is **equidistant** (same distance) from A and B.

Step 1
Draw AB = 7 cm.

Step 2
With the compass opened more than half of AB, draw arcs from A above and below AB.

Step 3
With the **same** opening, draw arcs from B so that they cut the first arcs.

Step 4
Join the two intersection points; the line meets AB at its midpoint M and AM = MB = 3.5 cm.

The arcs of equal radius from A and B meet at points equally far from A and B, which is why the line through them is the set of **all equidistant points**. This idea is used to find the centre of a circle through two points, or the location equally far from two villages.

8.4 Using Properties to Identify Shapes

- Diagonals equal and bisecting each other → **rectangle**.
- Diagonals equal, bisecting and perpendicular → **square**.
- A rectangle with a diagonal of 10 cm has the other diagonal also 10 cm.
- A tile pattern of squares of side 10 cm, 6 across and 4 down, forms a 60 cm × 40 cm rectangle: area 2,400 cm^2, perimeter 200 cm.
- Rectangles of whole-number sides with perimeter 16 cm: (1, 7), (2, 6), (3, 5) and the square (4, 4).

IMPORTANT

A construction must be **accurate**: keep the compass opening fixed while drawing arcs, draw thin light arcs (do not erase construction marks) and **verify** by measuring.

Quick Revision: Likely Exam Questions

Likely question	Model answer
State two properties of a rectangle.	Opposite sides are equal and parallel; every angle is 90°; diagonals are equal.
State two properties of the diagonals of a square.	They are equal and bisect each other at right angles.

Likely question	Model answer
Define an arc.	A part of the circumference of a circle.
Difference between a chord and a diameter?	A chord joins any two points on the circle; a diameter is a chord through the centre.
What is meant by equidistant?	At the same distance from the given points.
Why is every square a rectangle but not conversely?	A square has all rectangle properties plus equal sides.
Steps for a perpendicular bisector of AB?	Equal arcs from A and B on both sides; join the intersection points.
Quadrilateral with equal diagonals bisecting at 90°?	A square.
What are concentric circles?	Circles with the same centre and different radii.
How many right angles does a square have?	Four.

★ KEY TAKEAWAYS

- A circle's **radius, diameter, chord and arc** come from the compass; diameter = 2 × radius.
- A **rectangle** has equal diagonals; a **square** has equal diagonals that meet at 90°.
- Every square is a rectangle, but a rectangle need not be a square.
- The **perpendicular bisector** cuts a segment at its midpoint at 90° and holds all points equidistant from the ends.
- Keep the compass **opening fixed** and **verify** the finished construction by measuring.

CHAPTER 09 · MATHS (PART 2)

Symmetry

Symmetry is balance. A figure that can be folded so that both halves coincide has line symmetry; a figure that fits on itself after a turn has rotational symmetry. This chapter finds lines of symmetry, reflects figures in a mirror line and finds the order and angle of rotational symmetry.

Big Questions

- What is line symmetry and how many lines can a figure have?
- How is a figure reflected in a mirror line?
- What are the centre, order and angle of rotational symmetry?

9.1 Line Symmetry

DEFINITION

A figure has **line symmetry** if a line divides it into two halves that **coincide exactly** when folded along it. The line is the **line (axis) of symmetry** or mirror line.

Figure	Lines of symmetry	Where
Equilateral triangle	3	from each vertex to the midpoint of the opposite side
Isosceles triangle	1	from the apex to the midpoint of the base
Scalene triangle	0	no two sides are equal
Rectangle	2	joining midpoints of opposite sides (not the diagonals)
Square	4	2 diagonals and 2 midpoint lines
Rhombus	2	along the two diagonals
Regular pentagon / hexagon	5 / 6	as many lines as sides
Circle	infinitely many	every diameter

Letters: vertical axis: A, H, I, M, O, T, U, V, W, X, Y; horizontal axis: B, C, D, E, H, I, K, O, X; none: F, G, J, L, N, P, Q, R, S, Z. A parallelogram (not a rectangle) has **no** line of symmetry.

9.2 Reflection

DEFINITION

The **reflection** (mirror image) of a figure in a line is formed so that each point and its image are at **equal distances** from the line, on opposite sides, and the line joining them is **perpendicular** to the mirror line.

Step 1
From each important point draw a perpendicular to the mirror line.

Step 2
Extend it by the same distance on the other side.

Step 3
Mark the image points and join them in the same order.

In a reflection the figure is **reversed left to right**: the letter F in a mirror looks like a backward F. A figure with line symmetry is its own mirror image.

9.3 Rotational Symmetry

DEFINITION

A figure has **rotational symmetry** if it looks exactly the same after being turned through an angle less than a full turn about a fixed point, the **centre of rotation**. The number of times it fits on itself in one full turn is the **order** of rotational symmetry.

FORMULA

Smallest angle of rotation = 360° ÷ order. Regular hexagon: 360° ÷ 6 = 60°. Order 3 gives 120°. Square: order 4, angle 90°.

Figure	Order	Smallest angle
Equilateral triangle	3	120°
Square	4	90°
Rectangle	2	180°
Regular hexagon	6	60°
Letters H, N, S, X, Z	2	180°
Circle	infinite	any angle

NOTE

Every figure fits on itself after a full turn (order 1), but we say it has rotational symmetry only when the order is **2 or more**. A scalene triangle has order 1 only.

9.4 Symmetry in Life and Art

- **Architecture:** the Taj Mahal, temples and arches use line symmetry for balance.
- **Art and design:** rangoli, kolam, floor tiles, flags and logos use both kinds.
- **Nature:** butterflies and leaves (line symmetry), flowers and starfish (rotational symmetry).

A pentagon with all its lines of symmetry: five lines, each from a vertex to the midpoint of the opposite side. A rhombus and a rectangle both have two lines of symmetry, but only the rectangle's lines join midpoints of sides.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is line symmetry? Give an example.	A figure that can be folded along a line so both halves coincide, e.g. the letter A.
Lines of symmetry of an equilateral triangle, rectangle, circle?	3, 2 and infinitely many.

Likely question	Model answer
Angle of rotational symmetry of a regular hexagon?	360° ÷ 6 = 60°.
Order of rotational symmetry of the letter H?	2.
Order and angle for a square?	Order 4; angle 90°.
Which quadrilaterals have 2 lines of symmetry?	Rectangle and rhombus.
Why does a scalene triangle have no line of symmetry?	No two sides are equal, so no fold matches the halves.
What is the centre of rotation?	The fixed point about which the figure is turned.
State a property of reflection.	A point and its image are at equal distances from the mirror line.
Difference between line and rotational symmetry?	Line symmetry uses folding; rotational symmetry uses turning about a point.

★ KEY TAKEAWAYS

- **Line symmetry:** the figure folds onto itself; the fold is the **mirror line**.
- Equilateral triangle 3, square 4, rectangle 2, rhombus 2, regular n-gon **n** lines, circle infinite.
- **Reflection:** the image is as far behind the mirror as the object is in front, and is reversed.
- **Rotational symmetry:** order = number of matches in a full turn; **angle = 360° ÷ order**.
- A figure may have line symmetry, rotational symmetry, both or neither.

CHAPTER 10 · MATHS (PART 2)

The Other Side of Zero

Debts, temperatures below freezing and depths below sea level need numbers on the other side of zero. This chapter introduces negative numbers and integers, places them on the number line, compares them, and learns the rules for adding and subtracting integers with real-life examples.

Big Questions

- What are integers and how are they shown on a number line?
- How do we compare, add and subtract integers?
- Where do negative numbers occur in real life?

10.1 Negative Numbers and Integers

Numbers to the **left of zero** on the number line are **negative**; those to the right are **positive**. The **integers** are …, −3, −2, −1, 0, 1, 2, 3, …: the whole numbers together with their negatives. Zero is **neither positive nor negative**.

Situation	Positive integer	Negative integer
Money	profit / fortune: +₹40	loss / debt: −₹40
Height	20 m above sea level: +20	20 m below sea level: −20
Temperature	5 °C above zero: +5	5 °C below zero: −5
Direction	8 steps east: +8	8 steps west: −8
Time	5 minutes after the start: +5	5 minutes before: −5

NOTE

The ancient Indian mathematician **Brahmagupta** (7th century) gave rules for **fortunes** (positive) and **debts** (negative), including zero. A debt of 5 added to a debt of 3 is a debt of 8: (−5) + (−3) = −8.

10.2 The Number Line, Order and Absolute Value

- On the number line the integers are equally spaced; **greater** means **further to the right**.
- Every positive integer is greater than every negative integer and than 0; 0 is greater than every negative integer.
- Among negatives the one **closer to zero is greater**: −2 > −6.
- Ascending order: −6, −2, 0, 3, 5.
- The **absolute value** |a| is the distance of a from 0, so |−15| = |15| = 15.
- The **opposite** (additive inverse) of 7 is −7; a number and its opposite add to 0.
- **Successor** = number + 1 and **predecessor** = number − 1; the successor of the greatest negative integer (−1) is 0.

10.3 Adding Integers

Case	Rule	Example
Same signs	add the numbers, keep the common sign	(−7) + (−5) = −12; 3 + 4 = 7
Different signs	subtract the smaller from the larger (ignoring signs); take the sign of the **larger** number	(−7) + 5 = −2; (−3) + 8 = 5
Opposites	sum is 0	(−9) + 9 = 0

On the number line: to add a positive number, move **right**; to add a negative number, move **left**. −2 + 5: start at −2, move 5 right, reach **3**.

10.4 Subtracting Integers

FORMULA

a − b = a + (opposite of b). Subtracting a number is the same as adding its opposite.

Subtraction	Becomes	Answer
5 − 12	5 + (−12)	−7
(−3) − (−8)	(−3) + 8	5
(−6) − 4	(−6) + (−4)	−10
(−4) − (−9)	(−4) + 9	5
7 − (−3)	7 + 3	10

10.5 Word Problems

- A diver at −12 m goes 5 m up: −12 + 5 = **−7 m**.
- Profit ₹500 on Monday and loss ₹800 on Tuesday: 500 − 800 = **−₹300** (a net loss of ₹300).
- Temperature 8 °C falls 3 °C each hour for 5 hours: 8 − 15 = **−7 °C**.
- Hill 300 m high and pit −50 m deep: difference = 300 − (−50) = **350 m**.
- Ravi walks 8 steps east and 11 steps west: 8 − 11 = **−3**, i.e. 3 steps west of the start.
- The integer that added to −9 gives 4 is 4 − (−9) = **13**.

REMEMBER

The minus sign has two jobs: **subtraction** (7 − 3) and **negative sign** (−3). Brackets help: 7 + (−3). Do not confuse the **larger-looking** number with the **greater** one: −9 is smaller than −2.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What are integers? Give examples.	Whole numbers with their negatives: −3, −1, 0, 5.
Which is greater, −6 or −2?	−2, because it lies to the right on the number line.

Likely question	Model answer
Find (−8) + (−5).	−13.
Find (−4) − (−9).	−4 + 9 = 5.
Absolute value of −15?	15, its distance from zero.
Is zero positive?	No; zero is neither positive nor negative.
Write the integer for 20 m below sea level and for profit of ₹40.	−20 and +40.
Rule for adding integers with different signs?	Subtract the smaller from the larger and take the sign of the larger.
Evaluate 5 − 12.	−7.
Express a debt of 5 added to a debt of 3.	(−5) + (−3) = −8.

★ KEY TAKEAWAYS

- **Integers** = …, −2, −1, 0, 1, 2, …; zero is neither positive nor negative.
- On the number line **right is greater**; among negatives the one nearer zero is greater.
- **Same signs:** add and keep the sign; **different signs:** subtract and take the sign of the larger.
- **a − b = a + (−b)**: subtraction is addition of the opposite.
- **Brahmagupta** wrote the first rules for negative numbers as debts and fortunes.